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United States Patent Application Publication

20250284150

Kind Code

A1

Publication Date

September 11, 2025

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OPTICAL PHASE SHIFTER

Abstract

An optical phase shifter includes a core portion and a slab portion. The core portion has a first convex portion of a first conductivity type and a second convex portion of a second conductivity type. The slab portion has a first slab portion of a first conductivity type and a second slab portion of a second conductivity type. The first slab portion is electrically connected to the first convex portion, and the second slab portion is electrically connected to the second convex portion. Within a predetermined region where a PN junction is formed between the slab portion and the core portion, the second convex portion is arranged on both sides of the core portion adjacent to the slab portion, and the first slab portion is arranged on sides of the core portion.

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Family ID: 96929734

Appl. No.: 19/037412

Filed: January 27, 2025

Foreign Application Priority Data

JP	2024-034000	Mar. 06, 2024
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Publication Classification

Int. Cl.: G02F1/025 (20060101)

U.S. Cl.:

CPC G02F1/025 (20130101); G02F2201/063 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is based on Japanese Patent Application No. 2024-034000 filed on Mar. 6, 2024, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an optical phase shifter.

BACKGROUND

[0003] An optical phase shifter has a PN junction. Specifically, the optical phase shifter has a waveguide that extends in one direction as extension direction. The waveguide includes a core portion through which light propagates. The core portion is interposed between a first slab portion and a second slab portion, each of which having a thickness thinner than that of the core portion.

SUMMARY

[0004] According to one aspect of the present disclosure, an optical phase shifter includes: a core portion extending in an extending direction and having a thickness direction intersecting with the extending direction; and a slab portion arranged on sides of the core portion. A length of the slab portion in the thickness direction is shorter than that of the core portion. The core portion has a first convex portion of a first conductivity type and a second convex portion of a second conductivity type. The slab portion has a first slab portion of a first conductivity type and a second slab portion of a second conductivity type. The first slab portion is electrically connected to the first convex portion, and the second slab portion is electrically connected to the second convex portion. The first convex portion and the second convex portion are arranged to form a PN junction. The slab portion is made of an ion implantation layer, and is located to include a portion that forms the PN junction with the core portion, the PN junction being formed between the slab portion and the core portion within a predetermined region. The second convex portion is arranged on both sides of the core portion adjacent to the slab portion within the predetermined region. The first slab portion is arranged on sides of the core portion within the predetermined region.

Description

BRIEF DESCRIPTION OF DRAWING

[0005] FIG. 1 is a plan view of an optical phase shifter according to a first embodiment.

[0006] FIG. 2 is a cross-sectional view taken along line II-II in FIG. 1.

[0007] FIG. 3 is a cross-sectional view taken along line III-III in FIG. 1.

[0008] FIG. 4 is a cross-sectional view taken along line IV-IV in FIG. 1.

[0009] FIG. 5 is a schematic diagram showing positions of a first slab portion and a second slab portion when misalignment occurs in an optical phase shifter of a comparative example and the optical phase shifter of the first embodiment.

[0010] FIG. 6 is a diagram showing a relationship between a change in the amount of phase modulation and a misalignment of the first slab portion and the second slab portion.

[0011] FIG. 7 is a plan view of an optical phase shifter according to a modification of the first embodiment.

[0012] FIG. 8 is a plan view of an optical phase shifter according to a second embodiment.

[0013] FIG. 9 is a plan view of an optical phase shifter according to a third embodiment.

[0014] FIG. 10 is a cross-sectional view taken along line X-X in FIG. 9.

[0015] FIG. 11 is a cross-sectional view taken along line XI-XI in FIG. 9.

[0016] FIG. 12 is a cross-sectional view taken along line XII-XII in FIG. 9.

[0017] FIG. 13 is a cross-sectional view taken along line XIII-XIII in FIG. 9.

DETAILED DESCRIPTION

[0018] An optical phase shifter has a PN junction. Specifically, the optical phase shifter has a waveguide that extends in one direction as extension direction. The waveguide includes a core portion through which light propagates, and a first slab portion and a second slab portion between which the core portion is arranged. Each of the first slab portion and the second slab portion has a thickness thinner than that of the core portion.

[0019] More specifically, the core portion has an N-type first convex portion and a P-type second convex portion. The first convex portion has a generally L-shape in a plan view, formed of an extension portion extending along the extension direction, and a lateral portion extending from an end of the extension portion along a lateral direction that crosses the extension direction. The second convex portion has a generally L-shape in a plan view, formed of an extension portion extending along the extension direction, and a lateral portion extending from an end of the extension portion along a lateral direction that crosses the extension direction. In the core portion, the first convex portion and the second convex portion, each of which being the generally L-shape, are fitted together. Specifically, the extension portion of the first convex portion and the extension portion of the second convex portion are adjacent to each other. The lateral portion of the first convex portion is connected to the extension portion of the second convex portion, and the lateral portion of the second convex portion is connected to the extension portion of the first convex portion.

[0020] The first slab portion is of an N⁺ type having a higher impurity concentration than the N-type of the core portion. The first slab portion is arranged on the opposite side of the extension portion of the first convex portion through the extension portion of the first convex portion, and is connected to the extension portion of the second convex portion and the lateral portion of the first convex portion. The second slab portion is of a P⁺ type having a higher impurity concentration than the P-type of the core portion. The second slab portion is arranged on the opposite side of the extension portion of the second convex portion through the extension portion of the first convex portion, and is connected to the extension portion of the first convex portion and the lateral portion of the second convex portion.

[0021] In the optical phase shifter, the core portion, the first slab portion, and the second slab portion are ion-implanted layers formed by ion implantation.

[0022] In the optical phase shifter as described above, there is a possibility that the first slab portion and the second slab portion may be misaligned in the lateral direction with respect to the core portion due to a mask misalignment or the like during ion implantation. If the first slab portion and the second slab portion are misaligned in the lateral direction, there is a possibility that the change in the optical modulation efficiency will become large.

[0023] The present disclosure provides an optical phase shifter capable of reducing changes in modulation efficiency.

[0024] According to one aspect of the present disclosure, an optical phase shifter includes a core portion extending in one direction as extension direction, and a slab portion disposed at sides of the core portion. The slab portion has a thickness in a thickness direction intersecting the extension direction, and the thickness of the slab portion is shorter than that of the core portion. The core portion has a first convex portion of a first conductivity type and a second convex portion of a second conductivity type. The first convex portion and the second convex portion are disposed in a state of forming a PN junction. The slab portion is configured by an ion implantation layer. The slab portion has a first slab portion of a first conductivity type and is electrically connected to the first convex portion, and a second slab portion of a second conductivity type and is electrically connected to the second convex portion. The slab portion is arranged to include a portion which forms a PN junction with the core portion. Within a predetermined region where the PN junction is formed between the slab portion and the core portion, the second convex portions are arranged on both sides of the core portion adjacent to the slab portion, and the core portion is interposed

between the first slab portions.

[0025] Accordingly, the core portion is interposed between the first slab portions within the predetermined region where the PN junction is formed between the core portion and the slab portion. Therefore, even if the first slab portion and the second slab portion are misaligned in the lateral direction intersecting the extension direction and the thickness direction, this misalignment can be easily offset. Therefore, it is possible to suppress a large change in the modulation efficiency.

[0026] Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. In the following embodiments, the same or equivalent parts are denoted by the same reference numerals.

First Embodiment

[0027] An optical phase shifter of the first embodiment will be described with reference to FIGS. 1 to 4. The optical phase shifter of this embodiment is suitable for use in, for example, optical fiber communication devices.

[0028] The optical phase shifter of this embodiment includes a semiconductor substrate **10** made of a Silicon On Insulator (SOI) substrate in which a support substrate **11**, an insulating layer **12**, and an active layer **13** are stacked. In this embodiment, the support substrate **11** is made of silicon or the like. The insulating layer **12** is made of an oxide film or the like, and the active layer **13** is made of silicon or the like.

[0029] Hereinafter, one direction in the surface direction of the semiconductor substrate **10** will be referred to as X direction. A direction perpendicular to the X direction and along the surface direction of the semiconductor substrate **10** will be referred to as Y direction. A direction perpendicular to the X direction and the Y direction will be referred to as Z direction. In FIG. 1, the left-right direction on the paper surface is the X direction. The up-down direction on the paper surface is the Y direction, and a direction perpendicular to the paper surface is the Z direction. The Z direction is defined along a normal to the surface direction of the semiconductor substrate **10**, and can also be said to be a thickness direction of the semiconductor substrate **10**. Furthermore, the Z direction is defined along the stacking direction of the support substrate **11**, the insulating layer **12**, and the active layer **13**. In the following description, the X direction is also referred to the width direction. The length in the X direction is simply referred to as the width, and the length in the Z direction is simply referred to as the thickness.

[0030] The active layer **13** includes a core portion **20** doped with impurities, a first slab portion **31**, a second slab portion **32**, a first contact portion **41**, a second contact portion **42**, and the like so as to form a waveguide. The core portion **20**, the first slab portion **31**, the second slab portion **32**, the first contact portion **41**, and the second contact portion **42** are formed of ion-implanted layers into which impurity ions are implanted. Moreover, the active layer **13** of this embodiment includes a non-doped layer **60** that is not doped with impurities. Furthermore, the active layer **13** is provided with a first electrode portion **51**, a second electrode portion **52**, and the like, as described below.

[0031] The core portion **20** extends along the Y direction in the surface direction of the semiconductor substrate **10**, and has an N-type first convex portion **21** and a P-type second convex portion **22**. In this embodiment, the first convex portion **21** and the second convex portion **22** have approximately the same impurity concentration. In this embodiment, the Y direction corresponds to an extending direction of the core portion **20**, and light propagates along the Y direction.

[0032] The first convex portion **21** has an approximately T-shape formed by a width portion **21a** having the same width as the width of the core portion **20**, and an extension portion **21b** extending in the Y direction from approximately the center of the width portion **21a** in the X direction. The second convex portion **22** has an approximately U-shape formed by a width portion **22a** having the same width as the width of the core portion **20**, and an extension portion **22b** extending in the Y direction from both ends of the width portion **22a** in the X direction.

[0033] The core portion **20** is configured so that the first convex portion **21** and the second convex

portion **22** are fitted together. Specifically, the first convex portion **21** and the second convex portion **22** are fitted together such that the extension portion **21b** of the first convex portion **21** is interposed between the extension portions **22b** of the second convex portion **22**. In the core portion **20**, the P-type, the N-type, and the P-type are arranged in this order in the X direction within an area where the extension portion **21b** is interposed between the extension portions **22b**. That is, both ends of the core portion **20** in the X direction are the P-type in the area where the extension portion **21b** of the first convex portion **21** is interposed between the extension portions **22b** of the second convex portion **22**.

[0034] In this embodiment, a region where one first convex portion **21** and one second convex portion **22** are fitted together is referred to as one cell region SR. The cell region SR also includes the first slab portion **31** and the second slab portion **32** located adjacent to the first convex portion **21** and the second convex portion **22** fitted together in the X direction. The optical phase shifter is configured by repeatedly arranging the cell regions SR in the Y direction. In this embodiment, a region where only the first convex portion **21** is arranged as the core portion **20** in the X direction is described as a first extension region ER1. A region where the first convex portion **21** and the second convex portion **22** are arranged as the core portion **20** is described as a second extension region ER2. A region where only the second convex portion **22** is arranged as the core portion **20** is described as a third extension region ER3. In FIG. 1, the cell region SR is divided into the first extension region ER1, the second extension region ER2, and a third extension region ER3 from the bottom side of FIG. 1. FIG. 2 is a cross-sectional view of the first extension region ER1. FIG. 3 is a cross-sectional view of the second extension region ER2. FIG. 4 is a cross-sectional view of the third extension region ER3. A region located on one side of the core portion **20** in the X direction will be described as a first width region WR1, and a region located on the other side of the core portion **20** in the X direction will be described as a second width region WR2. In FIG. 1, a portion located to the left of the core portion **20** on the paper surface is the first width region WR1, and a portion located to the right of the core portion **20** on the paper surface is the second width region WR2.

[0035] The first slab portion **31** and the second slab portion **32** are thinner than the core portion **20**. Therefore, the first slab portion **31** and the second slab portion **32** can also be said to form a recess for the waveguide.

[0036] The first slab portion **31** is of N+ type and has a higher impurity concentration than the first convex portion **21** of the core portion **20**. In the first extension region ER1, the first slab portion **31** is arranged in the first width region WR1 and the second width region WR2 with the core portion **20** interposed therebetween, and is connected to the width portion **21a** in the core portion **20**.

[0037] In the second extension region ER2, the first slab portion **31** is arranged in the first width region WR1 and the second width region WR2 with the core portion **20** interposed therebetween. That is, in the second extension region ER2, the core portion **20** is interposed between the first slab portions **31** of a conductivity type (i.e., N-type) different from the conductivity type (i.e., P-type) of both ends in the X direction. In other words, in the second extension region ER2, the N-type first slab portion **31** is arranged to form a PN junction with each of the extension portions **22b** (i.e., the P-type layer) arranged on both ends in the X direction of the core portion **20**. However, in this embodiment, taking into consideration mask misalignment when forming the first slab portion **31** etc. by ion implantation, the non-doped layer **60** is interposed between the first slab portion **31** and the core portion **20** within the second extension region ER2.

[0038] In the third extension region ER3, the first slab portion **31** is disposed in the first width region WR1 to form a PN junction with the width portion **22a**. However, in this embodiment, taking into consideration mask misalignment when forming the first slab portion **31** etc. by ion implantation, the non-doped layer **60** is interposed between the first slab portion **31** and the core portion **20** within the third extension region ER3.

[0039] The first slab portion **31** extends along the Y direction in the first width region WR1. The

first slab portions **31** formed in the first extension region **ER1**, the second extension region **ER2**, and the third extension region **ER3** are coupled and electrically connected with each other. In addition, the first slab portion **31** formed in the first extension region **ER1** and the second extension region **ER2** of the second width region **WR2** is electrically connected to the first slab portion **31** formed in the first width region **WR1** via the width portion **21a** of the first convex portion **21**.

[0040] The second slab portion **32** is of P+ type having a higher impurity concentration than the second convex portion **22** of the core portion **20**, and is disposed in the second width region **WR2**. More specifically, in the first extension region **ER1** and the second extension region **ER2**, the second slab portion **32** is disposed opposite to the core portion **20** through the first slab portion **31**. The second slab portion **32** is disposed in the third extension region **ER3** so as to be connected to the width portion **22a** of the second convex portion **22** in the core portion **20**.

[0041] The second slab portion **32** formed in the second width region **WR2** extends along the Y direction. The second slab portions **32** formed in the first extension region **ER1**, the second extension region **ER2**, and the third extension region **ER3** are coupled and electrically connected with each other.

[0042] The first contact portion **41** is of N++ type, with an impurity concentration higher than that of the first slab portion **31**. The first contact portion **41** extends along the Y direction in the first width region **WR1**. The first contact portion **41** is positioned opposite to the core portion **20** through the first slab portion **31**, and is connected to the first slab portion **31**.

[0043] The second contact portion **42** is of P++ type, with an impurity concentration higher than that of the second slab portion **32**. The second contact portion **42** extends along the Y direction in the second width region **WR2**. The second contact portion **42** is positioned opposite to the core portion **20** through the second slab portion **32**, and is connected to the second slab portion **32**.

[0044] In this embodiment, the first contact portion **41** and the second contact portion **42** have the same thickness as the core portion **20**.

[0045] The first electrode portion **51** is disposed on the first contact portion **41** and electrically connected to the first contact portion **41**. The first electrode portion **51** extends along the Y direction in the same manner as the first contact portion **41**. The second electrode portion **52** is disposed on the second contact portion **42** and electrically connected to the second contact portion **42**. The second electrode portion **52** extends along the Y direction in the same manner as the second contact portion **42**.

[0046] The above is the basic configuration of the optical phase shifter in this embodiment. In the present embodiment, the N-type corresponds to a first conductivity type, and the P-type corresponds to a second conductivity type. Next, the operation and effects of the optical phase shifter will be described.

[0047] In the optical phase shifter of this embodiment, in the second extension region **ER2**, the PN junction is formed between the extension portion **22b** of the second convex portion **22** and the first slab portion **31** in the first width region **WR1**, and between the extension portion **22b** of the second convex portion **22** and the first slab portion **31** in the second width region **WR2**. In the second extension region **ER2**, the PN junction is formed between the extension portion **21b** of the first convex portion **21** and each of the two extension portions **22b** of the second convex portion **22**. In the third extension region **ER3**, the PN junction is formed between the width portion **22a** of the second convex portion **22** and the first slab portion **31** of the first width region **WR1**.

[0048] In the optical phase shifter, light propagates along the extension direction of the core portion **20** (that is, the Y direction). At this time, a higher voltage is applied to the first contact portion **41** than to the second contact portion **42**, so that a depletion layer expands at each of the PN junctions, and the carrier density decreases, thereby modulating the phase of the light. Hereinafter, application of a higher voltage to the first contact portion **41** than to the second contact portion **42** (i.e., application of a higher voltage to the first convex portion **21** than to the second convex portion **22**) will be simply referred to as application of a reverse bias voltage.

[0049] The optical phase shifter of this embodiment is manufactured as follows.

[0050] First, the semiconductor substrate **10**, which is an SOI substrate, is prepared. Next, etching is performed using a mask to thin portions of the active layer **13** to be the first slab portion **31** and the second slab portion **32**. Then, N-type impurities and P-type impurities are appropriately ion-implanted using a mask to form the core portion **20** having the first convex portion **21** and the second convex portion **22**. Next, N-type impurities and P-type impurities are appropriately ion-implanted to form the first slab portion **31** and the second slab portion **32**.

[0051] Thereafter, N-type impurities and P-type impurities are appropriately ion-implanted to form the first contact portion **41** and the second contact portion **42**. Further, the first electrode portion **51** and the second electrode portion **52** are appropriately formed to manufacture the optical phase shifter.

[0052] When forming the first slab portion **31** and the second slab portion **32**, there is a possibility that the first slab portion **31** and the second slab portion **32** may be misaligned in the X direction due to a mask misalignment or the like. However, in this embodiment, since the optical phase shifter is configured as described above, even if the first slab portion **31** and the second slab portion **32** are misaligned in the X direction, the misalignment can be easily offset in the second extension region ER2, thereby restricting a large change in the modulation efficiency.

[0053] The change in modulation efficiency will be described in detail below in comparison with an optical phase shifter of a comparative example. In the following, the optical phase shifter of the comparative example shown in FIG. 5 will be described as examples of the comparative optical phase shifter. The cross-sectional view of the optical phase shifter in FIG. 5 shows a portion corresponding to the second extension region ER2 of this embodiment. In FIG. 5, the support substrate **11**, the insulating layer **12**, and the like are omitted in the optical phase shifter of the comparative example and the optical phase shifter of this embodiment.

[0054] First, in the optical phase shifter of the comparative example, as shown in FIG. 5, a core portion **20** has an N-type first convex portion **21** and a P-type second convex portion **22** arranged side by side in the X direction. In the comparative example, a P+ type second slab portion **32** is arranged opposite to the second convex portion **22** through the first convex portion **21**, and an N+ type first slab portion **31** is arranged opposite to the first convex portion **21** through the second convex portion **22**. In the comparative example, in a cross-section different from that of FIG. 5, the first convex portion **21** and the first slab portion **31** are electrically connected, and the second convex portion **22** and the second slab portion **32** are electrically connected.

[0055] FIG. 5 illustrates a no misalignment case in which the non-doped layer **60** having a desired width is disposed between the core portion **20** and the first slab portion **31** and between the core portion **20** and the second slab portion **32**.

[0056] In FIG. 5, in the comparative example, the misalignment is Min when the first slab portion **31** and the second slab portion **32** are misaligned by the width of the non-doped layer **60** toward the core portion **20**, in case where the misalignment is zero is taken as a reference. That is, in the comparative example, a positional misalignment in which the first slab portion **31** is in contact with the second convex portion **22** and the second slab portion **32** is in contact with the first convex portion **21** is shown as misalignment Min. In FIG. 5, in the case of misalignment Min, the first slab portion **31** is misaligned to the right side, and the second slab portion **32** is misaligned to the left side, in the optical phase shifter of the comparative example. According to the first embodiment, the first slab portion **31** and the second slab portion **32** are misaligned in the same direction as the first slab portion **31** and the second slab portion **32** of the comparative example, as the misalignment Min. In this case, in the present embodiment, the first slab portion **31** is shifted to the right side and the second slab portion **32** is shifted to the left side.

[0057] In FIG. 5, in the comparative example, when the misalignment is zero is taken as the reference, the case where the first slab portion **31** and the second slab portion **32** are misaligned by the width of the non-doped layer **60** away from the core portion **20** is shown as misalignment Max.

In FIG. 5, the misalignment Max refers to the case where the first slab portion **31** is misaligned to the left side and the second slab portion **32** is misaligned to the right side in the comparative example. In the first embodiment, the misalignment Max is defined as the case where the first slab portion **31** and the second slab portion **32** are misaligned in the same direction as the case where the first slab portion **31** and the second slab portion **32** of the comparative example are misaligned. That is, in the optical phase shifter of this embodiment, the case where the first slab portion **31** is shifted to the left side and the second slab portion **32** is shifted to the right side is shown as the misalignment Max.

[0058] The inventors then conducted extensive research into a phase modulation amount $V_{sub.TT}$ when misalignment occurs as shown in FIG. 5 for the optical phase shifter of the comparative example and the optical phase shifter of this embodiment, and obtained the results shown in FIG. 6.

[0059] As shown in FIG. 6, if the phase modulation amount $V_{sub.TT}$ when the misalignment is zero is taken as the reference (i.e., the change in phase modulation amount $V_{sub.TT}$ is 1), it is confirmed that in the optical phase shifter of the comparative example, the phase modulation amount $V_{sub.TT}$ changes significantly due to misalignment. The phase modulation amount $V_{sub.TT}$ corresponds to an applied voltage required to change the phase of light by 180° , and it is preferable that the phase modulation amount $V_{sub.TT}$ to be smaller.

[0060] In the comparative example, when the misalignment is at Min, since the first slab portion **31** and the second slab portion **32** are close to the core portion **20**, it is confirmed that the phase modulation amount $V_{sub.TT}$ is smaller than the reference value. However, in the comparative example, when the misalignment is at Max, it is confirmed that the phase modulation amount $V_{sub.TT}$ becomes larger than the reference value. That is, it is confirmed that, in the comparative example, the phase modulation amount $V_{sub.TT}$ changes greatly between when the misalignment is at Min and when the misalignment is at Max.

[0061] In contrast, in the optical phase shifter of this embodiment, the core portion **20** is interposed between the first slab portions **31** of the same conductivity type. For this reason, in this embodiment, as shown in FIG. 5, when the misalignment is Min, the first slab portion **31** in the first width region WR1 is misaligned toward the core portion **20**, while the first slab portion **31** in the second width region WR2 is misaligned away from the core portion **20**. Therefore, the misalignment of the first slab portion **31** in the first width region WR1 and the misalignment of the first slab portion **31** in the second width region WR2 are offset.

[0062] Similarly, when the misalignment is Max, the first slab portion **31** in the first width region WR1 is misaligned away from the core portion **20**, while the first slab portion **31** in the second width region WR2 is misaligned toward the core portion **20**. Therefore, the misalignment of the first slab portion **31** in the first width region WR1 and the misalignment of the first slab portion **31** in the second width region WR2 are offset.

[0063] Therefore, in the optical phase shifter of this embodiment, the phase modulation amount $V_{sub.TT}$ changes to the smaller side when the misalignment occurs to any side in the X direction. Therefore, in this embodiment, as shown in FIG. 6, a large change in the phase modulation amount $V_{sub.TT}$ due to misalignment is suppressed.

[0064] In the optical phase shifter of this embodiment, when the misalignment is at Min, in the second width region WR2, the misalignment is reversed between the first slab portion **31** and the second slab portion **32**, and the width of the first slab portion **31** or the second slab portion **32** becomes shorter. Similarly, when the misalignment is at Max, in the second width region WR2, the misalignment is reversed between the first slab portion **31** and the second slab portion **32**, and the non-doped layer **60** is formed between the first slab portion **31** and the second slab portion **32**. However, there is no significant effect on the modulation efficiency even if the configuration is changed slightly since these portions do not mainly contribute to the light propagation.

[0065] Furthermore, in the optical phase shifter of this embodiment, the impurity concentration of the first slab portion **31** is made higher than the impurity concentration of the first convex portion

21. Therefore, if misalignment causes the first slab portion 31 to reach the inside of the core portion 20, the loss of light becomes large. Therefore, in the first slab portion 31, it is preferable that the width of the non-doped layer 60 is adjusted not to reach the inside of the core portion 20 even if misalignment occurs.

[0066] According to the present embodiment, in the second extension region ER2 where a PN junction is formed between the core portion 20 and the slab portion 31, 32, the core portion 20 is interposed between the first slab portions 31. Therefore, even if the first slab portion 31 and the second slab portion 32 are misaligned in the X direction, this misalignment can be easily offset, making it possible to suppress a large change in modulation efficiency.

[0067] (1) In this embodiment, in the second extension region ER2, the core portion 20 has the first convex portion 21 interposed between the second convex portions 22 in the X direction, and a PN junction is formed between the first convex portion 21 and the second convex portion 22. This makes it possible to increase the number of PN junctions and improve the modulation efficiency.

[0068] (2) In this embodiment, the core portion 20, the first slab portion 31, and the second slab portion 32 are repeatedly provided in the Y direction. That is, in this embodiment, the cell regions SR are repeatedly provided in the Y direction. This makes it easier to spread the depletion layer uniformly within the core portion 20, thereby improving the modulation efficiency.

Modification of First Embodiment

[0069] In the first embodiment, a P-type, an N-type, and a P-type are arranged in the X direction in the second extension region ER2 of the core portion 20. However, as shown in FIG. 7, the optical phase shifter may be configured such that an N-type, a P-type, and an N-type are arranged in the second extension region ER2. That is, in the first embodiment, the first convex portion 21, the first slab portion 31, and the first contact portion 41 may be P-type, and the second convex portion 22, the second slab portion 32, and the second contact portion 42 may be N-type. In other words, the first convex portion 21, the first slab portion 31, and the first contact portion 41 in the first embodiment may be interchanged with the second convex portion 22, the second slab portion 32, and the second contact portion 42. In this configuration, a higher voltage than that applied to the first contact portion 41 may be applied to the second contact portion 42. In this configuration, the P-type is a first conductivity type, and the N-type is a second conductivity type.

Second Embodiment

[0070] A second embodiment is described. In this embodiment, the number of PN junctions in the second extension region ER2 is increased compared to the first embodiment. The remaining configurations are similar to those of the first embodiment and will thus not be described repeatedly.

[0071] As shown in FIG. 8, in the optical phase shifter of this embodiment, the first convex portion 21 has the width portion 21a and two extension portions 21b extending from the width portion 21a along the Y direction. The two extension portions 21b are disposed apart from each other in the X direction.

[0072] The second convex portion 22 has the width portion 22a and three extension portions 22b extending from the width portion 22a along the Y direction. The three extension portions 22b are disposed apart from each other in the X direction.

[0073] The first convex portion 21 and the second convex portion 22 are arranged such that the two extension portions 21b and the three extension portions 22b mesh with each other. As a result, in the core portion 20 of the second extension region ER2, a P-type, an N-type, a P-type, an N-type, and a P-type are arranged in the X direction. Compared to the first embodiment, the number of PN junctions increases, resulting in an increase in the number of depletion layers. Therefore, the modulation efficiency can be improved.

[0074] In this embodiment, the number of the extension portions 21b of the first convex portion 21 and the extension portions 22b of the second convex portion 22 can be changed as appropriate. However, if the number of PN junctions is increased too much, the depletion layers may be

connected when a reverse bias voltage is applied, resulting in a decrease in the amount of phase modulation. For this reason, it is preferable that the number of PN junctions formed inside the core portion **20** is set so that the depletion layers are not in contact with each other when a reverse bias voltage is applied. For example, when the width of the core portion **20** is about 500 nm, it is preferable that four PN junctions (that is, depletion layers) are formed inside the core portion **20** as in this embodiment.

[0075] According to the present embodiment, in the second extension region ER **2** where a PN junction is formed between the core portion **20** and the slab portion **31**, **32**, the core portion **20** is interposed between the first slab portions **31**. Therefore, effects similar to those of the first embodiment can be obtained.

[0076] (1) In this embodiment, the second extension region ER**2** includes plural first convex portions **21** interposed between the second convex portions **22**. This makes it possible to increase the number of PN junctions and improve the modulation efficiency. However, in such a configuration, it is preferable that the number of PN junctions formed inside the core portion **20** is set to such that the depletion layers are not in contact with each other when a reverse bias voltage is applied.

Third Embodiment

[0077] A third embodiment will be described. In this embodiment, the configuration of the core portion **20** is changed from that of the first embodiment. The remaining configurations are similar to those of the first embodiment and will thus not be described repeatedly.

[0078] As shown in FIGS. **9** to **13**, in the optical phase shifter of this embodiment, the core portion **20** is formed so that a PN junction is formed in the Z direction. FIG. **13** corresponds to a cross-section taken along line XIII-XIII in FIG. **9**, and shows the core portion **20** of one cell region SR.

[0079] Specifically, the first convex portion **21** in this embodiment has the width portion **21a** and the extension portion **21b** that is thinner than the width portion **21a** and extends in the Y direction. The width of the extension portion **21b** is the same as that of the width portion **21a**, and is configured to be thinner from the insulating layer **12**.

[0080] The second convex portion **22** has the same width as the core portion **20**, and has a thickness corresponding to the thinned portion of the extension portion **21b** of the first convex portion **21**.

[0081] The first convex portion **21** and the second convex portion **22** are arranged so as to form a PN junction in the Z direction (i.e., the XY plane). In this embodiment, a region in the X direction in which only the first convex portion **21** is arranged as the core portion **20** is defined as a first extension region ER**1**. In this embodiment, a region in which the first convex portion **21** and the second convex portion **22** are arranged as the core portion **20** interposed between the first slab portions **31** is defined as the second extension region ER**2**. In this embodiment, a region in which the first convex portion **21** and the second convex portion **22** are arranged as the core portion **20** and the second convex portion **22** is connected to the second slab portion **32** is referred to as the third extension region ER**3**. The first convex portion **21** and the second convex portion **22** are arranged such that a PN junction is formed in the second extension region ER**2** and the third extension region ER**3**.

[0082] The first slab portion **31** and the second slab portion **32** are arranged in the same manner as in the first embodiment. Therefore, in the second extension region ER**2**, the N-type first slab portion **31** is arranged in the first width region WR**1** and the second width region WR**2** so as to form a PN junction with the P-type second convex portion **22**. That is, in the second extension region ER**2**, the core portion **20** is interposed between the first slab portions **31** of the same conductivity type. In the third extension region ER**3**, the N-type first slab portion **31** is disposed in the first width region WR**1** so as to form a PN junction with the P-type second convex portion **22**. In the third extension region ER**3**, the P-type second slab portion **32** is disposed so as to be connected to the P-type second convex portion **22**. However, the first slab portion **31** and the

second slab portion **32** are made to have a thickness equal to or less than that of the second convex portion **22** arranged in the lower layer of the core portion **20**.

[0083] In such an optical phase shifter, in the second extension region ER2, a PN junction is formed between the second convex portion **22** and the first slab portion **31** of the first width region WR1, and between the second convex portion **22** and the first slab portion **31** of the second width region WR2. In the second extension region ER2, a PN junction is formed between the extension portion **21b** of the first convex portion **21** and the second convex portion **22**. In the third extension region ER3, a PN junction is formed between the second convex portion **22** and the first slab portion **31** of the first width region WR1. In the third extension region ER3, a PN junction is formed between the extension portion **21b** of the first convex portion **21** and the second convex portion **22**. Therefore, similarly to the first embodiment, the phase of light can be modulated by applying a reverse bias voltage.

[0084] Furthermore, in an optical phase shifter of this embodiment, when the first slab portion **31** and the second slab portion **32** are formed by ion implantation, there is a possibility that the first slab portion **31** and the second slab portion **32** may be misaligned in the X direction due to mask misalignment, and the like. However, in this embodiment, as in the first embodiment, in the second extension region ER2, the core portion **20** (i.e., the second convex portion **22**) is interposed between the first slab portions **31**. Therefore, when the first slab portion **31** and the second slab portion **32** are misaligned in the X direction, the misalignment can be easily offset, and a large change in the modulation efficiency can be suppressed.

[0085] According to the present embodiment, in the second extension region ER 2 where a PN junction is formed between the core portion **20** and the slab portion **31**, **32**, the core portion **20** is interposed between the first slab portions **31**. Therefore, effects similar to those of the first embodiment can be obtained.

[0086] (1) In this embodiment, a PN junction is formed in the portion where the first convex portion **21** is stacked on the second convex portion **22**, and the PN junction is formed in a direction along the XY plane. The core portion **20** is usually configured so that its width is longer than its thickness. Light tends to spread in the X direction. Therefore, by forming a PN junction in a direction along the XY plane, it is possible to easily improve the modulation efficiency.

Other Embodiments

[0087] Although the present disclosure has been described in accordance with the embodiments, it is understood that the present disclosure is not limited to such embodiments or structures. The present disclosure encompasses various modifications and variations within the scope of equivalents. In addition, various combinations and modes, and other combinations and modes including only one element, more elements, or less elements are also within the scope and idea of the present disclosure.

[0088] For example, in each of the embodiments, the optical phase shifter may be configured with one cell region SR.

[0089] In each of the embodiments, the first convex portion **21** and the second convex portion **22** may have different impurity concentrations.

[0090] In each of the embodiments, when the misalignment is zero, the core portion **20** may be in contact with the first slab portion **31** and the second slab portion **32**. However, as described in the first embodiment, if misalignment causes the first slab portion **31** or the second slab portion **32** to reach the inside of the core portion **20**, the loss of light becomes large. For this reason, it is preferable that the non-doped layer **60** is disposed when the misalignment is zero so that the first slab portion **31** or the second slab portion **32** does not reach the inside of the core portion **20** even if misalignment occurs.

[0091] In the second and third embodiments, similar to the modification of the first embodiment, the first convex portion **21**, the first slab portion **31**, and the first contact portion **41** may be the P-

type, and the second convex portion 22, the second slab portion 32, and the second contact portion 42 may be the N-type.

Claims

1. An optical phase shifter comprising: a core portion extending in an extending direction and having a thickness direction intersecting with the extending direction; and a slab portion arranged on sides of the core portion, a length of the slab portion in the thickness direction being shorter than that of the core portion, wherein the core portion has a first convex portion of a first conductivity type and a second convex portion of a second conductivity type, the slab portion has a first slab portion of a first conductivity type and a second slab portion of a second conductivity type, the first slab portion is electrically connected to the first convex portion, and the second slab portion is electrically connected to the second convex portion, the first convex portion and the second convex portion are arranged to form a PN junction, the slab portion is made of an ion implantation layer, and is located to include a portion that forms the PN junction with the core portion, the PN junction being formed between the slab portion and the core portion within a predetermined region, the second convex portion is arranged on both sides of the core portion adjacent to the slab portion within the predetermined region, and the first slab portion is arranged on sides of the core portion within the predetermined region.
 2. The optical phase shifter according to claim 1, wherein the second convex portion is arranged on sides of the first convex portion in a width direction intersecting the extending direction and the thickness direction within the predetermined region.
 3. The optical phase shifter according to claim 2, wherein the core portion has a plurality of sets in which the second convex portion is arranged on sides of the first convex portion in the width direction, and is configured such that depletion layers resulting from the PN junction between the first convex portion and the second convex portion are non-contact with each other.
 4. The optical phase shifter according to claim 1, wherein the first convex portion is disposed on an upper side of the second convex portion within the predetermined region where the PN junction is formed between the slab portion and the core portion.
 5. The optical phase shifter according to claim 1, wherein the predetermined region in which the PN junction is formed between the slab portion and the core portion is repeatedly arranged in the extending direction.
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